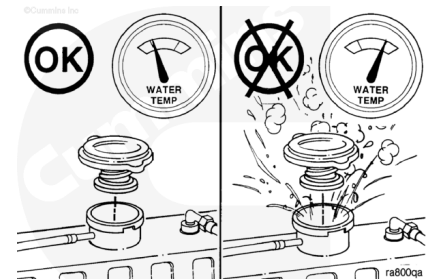


## Trainings ▾

### Drain

#### **⚠ WARNING ⚠**

**Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.**



#### **⚠ WARNING ⚠**

**Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.**

Remove the radiator cap after the engine is cool.

Open the petcocks on each turbocharger.

Open the draincock at the bottom of the radiator.

Remove the lower radiator hose.

Open the aftercooler core drain valves.

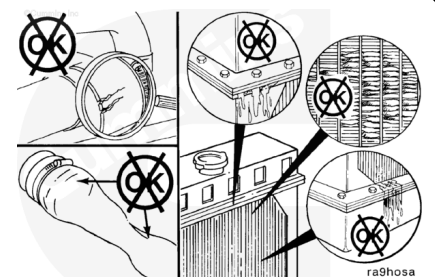
Drain the cooling system.

Do **not** allow the cooling system to dry out.

Check for damaged hoses and loose or damaged hose clamps. Replace as required.

Check the radiator for leaks, damage, and buildup of dirt.

Clean and replace as required.

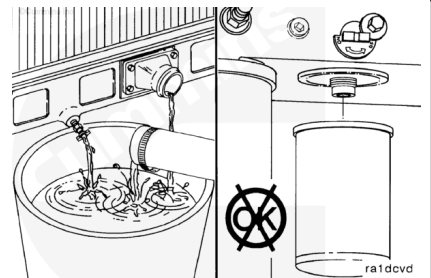


# Flush

RESTORE™ (cooling system cleaner) is a heavy-duty cooling system cleaner that removes corrosion products, silica gel, and other deposits. The performance of RESTORE™ is dependent on time, temperature, and concentration levels. An extremely scaled or flow-restricted system, for example, can require higher concentrations of cleaners, higher temperatures, longer cleaning times, or the use of RESTORE Plus™ (cooling system cleaner). Up to twice the recommended concentration levels of RESTORE™ can be used safely. RESTORE Plus™ **must** be used **only** at its recommended concentration level. Extremely scaled or fouled systems can require more than one cleaning.

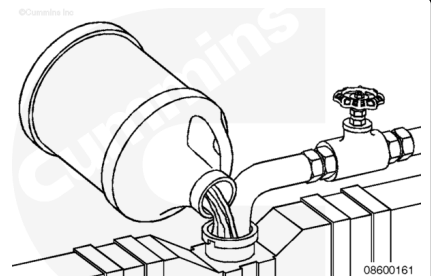


Do **not** remove the coolant filters.



## ⚠ CAUTION ⚠

**Fleetguard® Restore™ (cooling system cleaner) contains no antifreeze. Do not allow the cooling system to freeze during the cleaning operation.**



## ⚠ CAUTION ⚠

**Fill the cooling system slowly to avoid trapping air in the system. Air trapped inside the cooling system can lead to engine damage.**

## ⚠ CAUTION ⚠

**Be sure that the vent lines of the cooling system are routed correctly and are in good condition, with no kinks or pinches.**

Be sure the petcocks are open to avoid air entrapment in the cooling system.

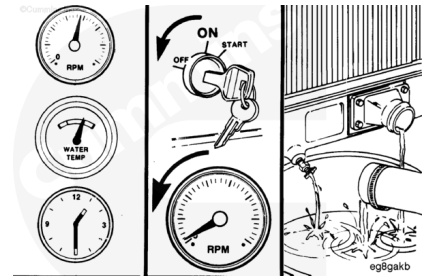
Immediately add 3.8 liters [1 gal] of Fleetguard® Restore™, or equivalent, for each 38 to 57 liters [10 to 15 gal] of cooling system capacity, and fill the system with plain water.

Turn the heater temperature switch to HIGH to allow maximum coolant flow through the heater core. The blower does **not** have to be on.

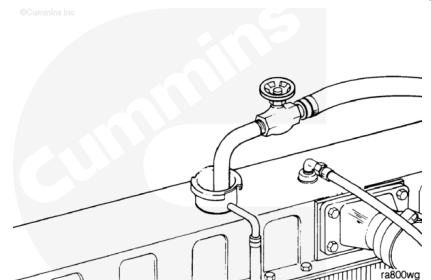
For correct filling instructions, see the Fill section below.

Operate the engine at a coolant temperature of at least 85°C [185°F] for 1 to 1 ½ hours.

Shut the engine OFF. Allow it to cool to 50°C [120°F]. Drain the cooling system.



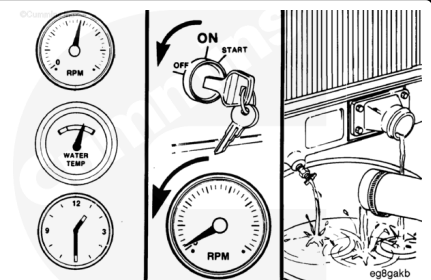
Fill the cooling system with clean water. See the Fill section below.



Operate the engine at high idle for 5 minutes with the coolant temperature above 85°C [185°F].

Shut the engine OFF. Allow it to cool to 50°C [120°F]. Drain the cooling system.

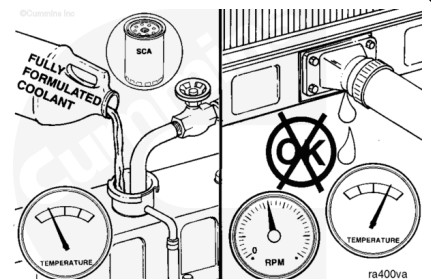
If the water drained is still dirty, the system **must** be flushed again until the water is clean.



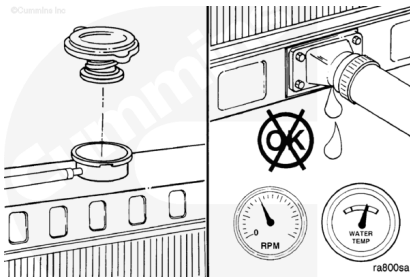
Install a new coolant filter.

Fill the cooling system with fully formulated coolant.

Use additional SCA to bring the coolant to the correct SCA concentration level. Refer to Procedure 018-004 in Section V. ([/qs3/pubsys2/xml/en/procedures/102/102-018-004.html](http://qs3/pubsys2/xml/en/procedures/102/102-018-004.html))



Install the pressure cap. Operate the engine until the coolant reaches a temperature of 70°C [158°F] and check for coolant leaks.



Measurements

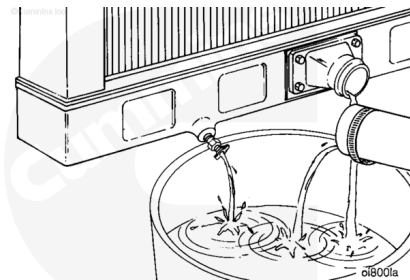
|                                              | kpa | psi |
|----------------------------------------------|-----|-----|
| Minimum Cooling System Pressure Cap Capacity | 48  | 7   |

Fill

Make sure the radiator draincocks and aftercooler core drain valves are closed.

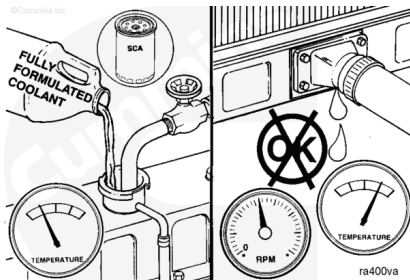
Open the cooling system venting petcocks.

Install the lower radiator hose(s). Refer to Procedure 008-045 in Section 7. (/qs3/pubsys2/xml/en/procedures/99/99-008-045.html)



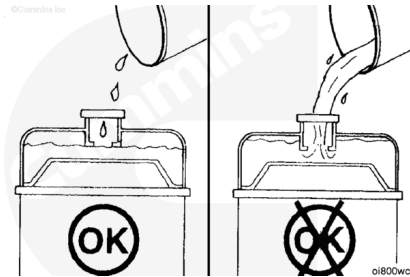
Use fully formulated coolant to fill the cooling system.

Use the correct units of SCA to obtain the correct cooling system protection. Refer to Procedure 018-004 in Section V. (/qs3/pubsys2/xml/en/procedures/102/102-018-004.html)



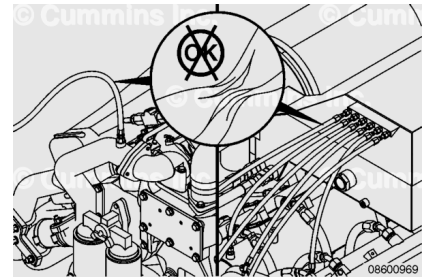
CAUTION

Fill the cooling system slowly to avoid trapping air in the cooling system. Air trapped in the system can lead to engine damage.



CAUTION

Be sure that the vent lines of the cooling system are routed correctly and are in good condition with no kinks or pinches.

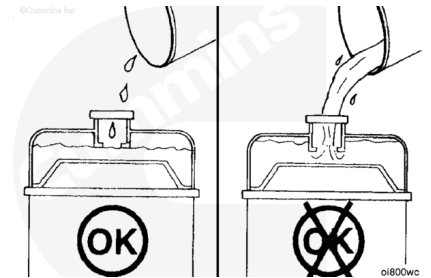


Begin filling the cooling system with coolant.

Fill until a continuous stream of coolant begins to flow from each opened petcock in the coolant system.

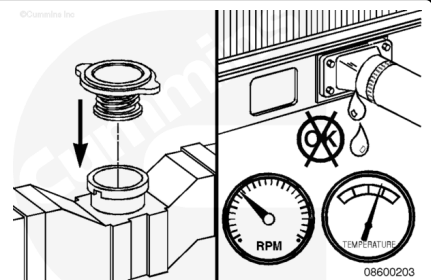
Close each petcock as the continuous stream of coolant begins to flow from that petcock.

Fill the system with the coolant to the bottom of the radiator fill neck.



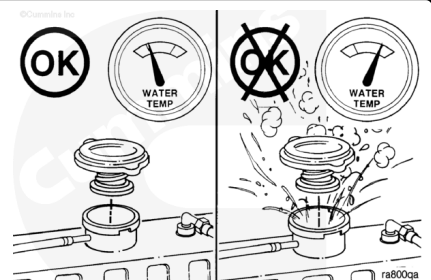
Install the cooling system pressure cap.

Operate the engine until the coolant reaches a temperature of 85°C [185°F]. Check for leaks.



### **⚠ WARNING ⚠**

**Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.**



Shut the engine off, and allow it to cool.

Check the coolant level.

